

epMotion Equimolar Pooling Helper

Quick User Guide

This tool calculates equimolar pooling volumes for sequencing libraries and generates Eppendorf epMotion worklists (DNA.csv and Buffer.csv).

1. Understanding the Two Main Sections

Pooling Section

Defines the desired final pool. Think of this as: *What should the final pool look like?*

- Platform: Illumina or ONT workflow presets
- Loading Preset: Predefined recommended settings
- Target Pool nM: Desired final concentration
- Total Volume (μL): Final pool volume
- Layout: One-Tube Pooling or Normalize

Sample Section

Describes the libraries you currently have. Think of this as: *What am I starting with?*

- Sample Name
- Concentration ($\text{ng}/\mu\text{L}$)
- Fragment Size (bp)
- Optional Direct nM Input

2. Typical Workflow

1. Select Platform and Preset.
2. Verify Target Pool nM and Total Volume.
3. Enter sample measurements.
4. Review the plate map.
5. Click Generate Worklists.
6. Review warnings and download files.

3. Buttons and Functions

Add Sample – Adds a new library row.

Import CSV – Imports sample information.

Export CSV/TXT – Saves current inputs.

Save Run – Saves the entire session as JSON.

Open Run – Reloads a saved session.

Blank Template – Downloads a sample-entry template.

New/Clear – Resets the run.

Generate Worklists – Performs calculations and creates output files.

4. Plate Map

The live plate map highlights occupied wells as samples are entered. Use it to verify rack loading before running the epMotion.

5. Output Files

DNA.csv – DNA transfer instructions.

Buffer.csv – Buffer transfer instructions.

Pooling Worksheet – Calculation summary.

Inputs File – Saved run settings and sample information.

6. Important Warnings

Minimum-volume warnings indicate transfers below reliable pipetting limits. Consider increasing pool volume or pre-diluting concentrated libraries. Always verify concentrations, fragment sizes, and plate positions before running the epMotion.